Marcus Justice Uy

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Summary

Detail-oriented Software Engineering graduate with strong foundational knowledge in data structures, algorithms, and object-oriented programming, complemented by hands-on experience developing and deploying cloud-based systems. Experienced in building backend services and APIs with a focus on scalability, performance, and clean code. Eager to contribute to real-world software systems in a collaborative, fast-paced environment while continuously expanding my skills as a quick and motivated learner.

Education

Bachelor of Engineering - Computer Engineering (Honours) 2021 – 2025
Toronto Metropolitan University (formerly Ryerson)
Dean's List: 2021/2022 | 2022/2023 | 2024/2025
Relevant Coursework: Data Structures & Algorithms, Software Architecture, Databases, Operating Systems, Computer Networks, Cloud Computing, Software Testing & Project Management

Professional Certifications

AWS Certified AI Practitioner November 2025 – November 2028
Issued by Amazon Web Services Training and Certification

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks: React.js, Bootstrap, Node.js, FastAPI, Flask, gRPC, PySpark, JUnit, TestNG, Selenium
Developer Tools: Git, Docker, Kubernetes, Azure Cloud, Google Cloud, Google Colab, Microsoft 365, RabbitMQ, Databricks, Gemini
Libraries: OpenCV, NumPy, Pandas, Scikit-learn, MongoDB (NoSQL), MySQL, Oracle SQL, OpenAI API
Concepts: Object-Oriented Programming (OOP), Data Structures and Algorithms, Distributed Systems, RESTful API Design, Multithreading, Networking, Operating Systems, Machine Learning Fundamentals

Work Experience

Sterling Information Technology Internship June 2021 – June 2023
- Worked with a 4-member technical team and 3 external clients to implement QoL improvements and new functionality for the ROSe Telehealth mobile app, delivering the project 20% ahead of schedule.
- Gathered and documented business and technical requirements, producing Business Requirements Documents (BRDs) and test result documentation to support development and QA processes.
- Conducted quality assurance testing, validating functionality against requirements and

- coordinating feedback with developers to ensure high client satisfaction.
- Designed user interface mock-ups and logic flows for a cybersecurity payment application focused on security upscaling
- Created and maintained technical documentation for company alerts, networks, servers, and rulesets, organizing system data into structured Excel reports.

Academic and Personal Projects

AI Automation for Processing Visa Statements May 2025 – Present

- Designed an AI automation workflow to automate PDF processing, transaction data extraction and parsing, classify transaction into spending categories, and inserts into Google Sheets
- Built a user-friendly interface that allows for seamless file uploads via drag and drop, that triggers the AI automation workflow upon user request
- Locally-hosted the application and n8n on my Raspberry Pi computer to ensure security when processing sensitive data, without relying on third-party services
- Reduced the time spent of manual data entry by over 90%, making it 25x faster
- **Technologies Used:** React, Vite, n8n, OpenAI API, Linux, Docker, nGinx

Banking App Finance Management Mar 2026 – Present

- Implemented a platform that connects to multiple bank accounts, displays transactions in real-time, allows users to transfer money to other users, and to manage their finances
- Utilized Appwrite and Zod for secure SSR authentication with proper validations
- Integrates with Plaid for bank account linking with respective balances and account details
- Seamless adaptability for desktop and mobile views, with intuitive layouts
- **Technologies Used:** Next.js, TypeScript, Tailwind CSS, Shadcn, Appwrite, Zod, Plaid, Dwolla

Ecommerce BI Dashboard Feb 2026 – Mar 2026

- Implemented a Medallion Architecture in Databricks, creating 6 schemas for analytics-ready data
- Processed 80k+ rows and 35+ attributes for ecommerce data using PySpark and SQL
- Ingested 100k+ historical order records from multiple CSV files representing 3 months of data
- Performed data cleaning and transformation including null values, anomaly detection, etc
- Built a Databricks dashboard to visualize business insights such as sales trends
- **Technologies Used:** Databricks, Python, PySpark, SQL, Jupyter Notebook, Gemini

AI Meal Plan Application Sep 2025 – Nov 2025

- Designed a 7-day meal planning system that utilizes OpenAI API to produce specific meals
- Reduced cost by 45% and improved latency by 67% through reduction in input and output tokens
- Built a responsive user interface, for user creation and user login
- Built a dashboard to keep track of past meal plans and the date that they were created
- Developed 6 backend API endpoints connected to NoSQL database, to keep track of users, profile information, and their generated meal plans
- **Technologies Used:** OpenAI API, Python, React.js, HTML, CSS, FastAPI, MongoDB, Bootstrap

Smart Waste Sorting System Sep 2024 – Apr 2025

- Collaborated with a team of 4 to develop and deploy a YOLOv11-powered Smart Waste Sorting System for real-time waste classification into garbage, recyclables, and compost
- Delivered a robust, low-latency system capable of operating on consumer-grade hardware under extended use conditions without major performance degradation
- Trained the model using over 50,000 images across three public datasets, with custom preprocessing and augmentation to improve detection performance
- Delivered a robust, low-latency system maintaining less than 200ms average inference time per image under extended use
- Improved overall detection accuracy by 74% and precision by 43% over the baseline model
- **Technologies Used:** Google Colab, Python, React.js, HTML, CSS, Flask, GIT